Exploratory Data Analysis (EDA)

Titanic Passenger Dataset

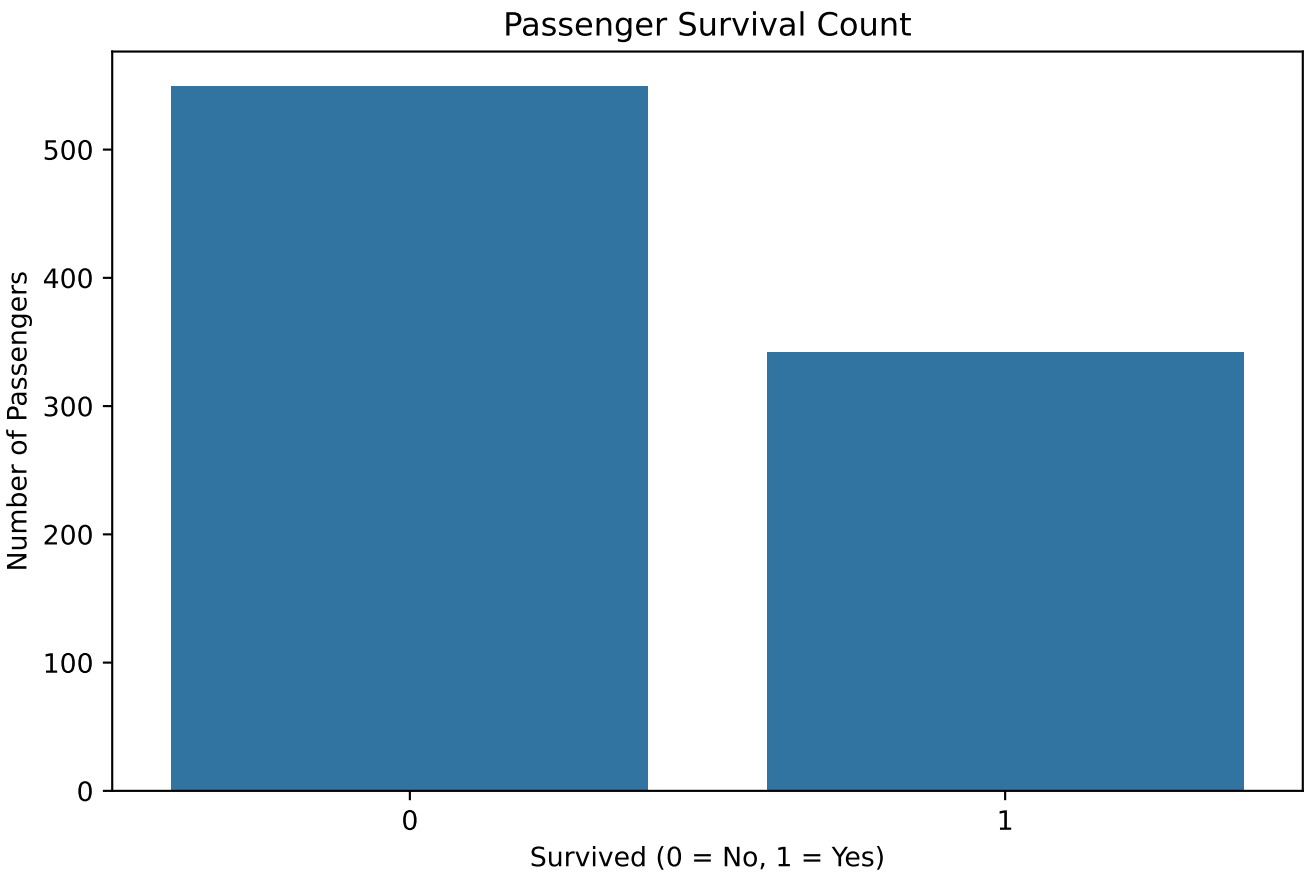
Dataset overview
- Total passenger records: 891
- Total columns: 12
- Passengers who survived: 342 (38.4%)
- Passengers who did not survive: 549 (61.6%)

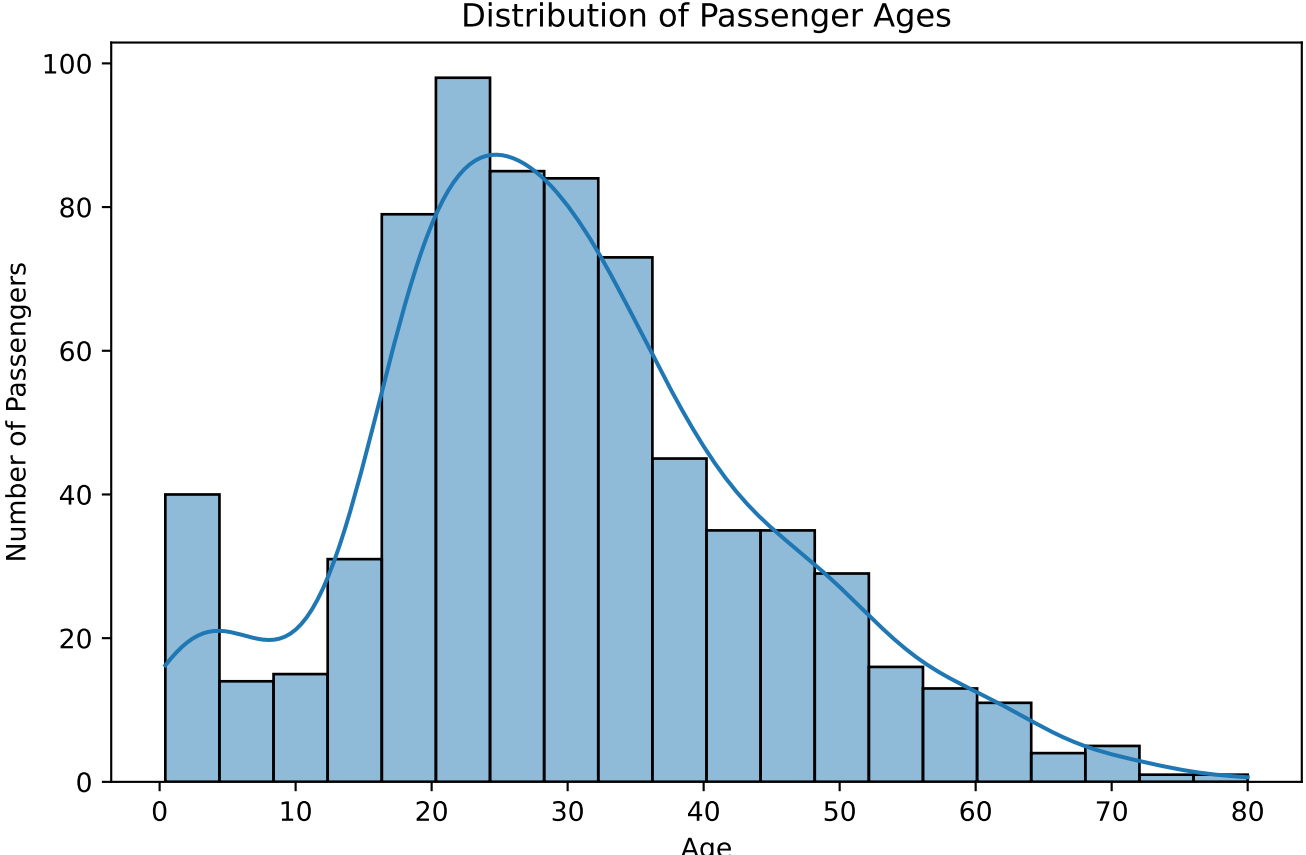
Missing values
- Age: 177
- Cabin: 687
- Embarked: 2

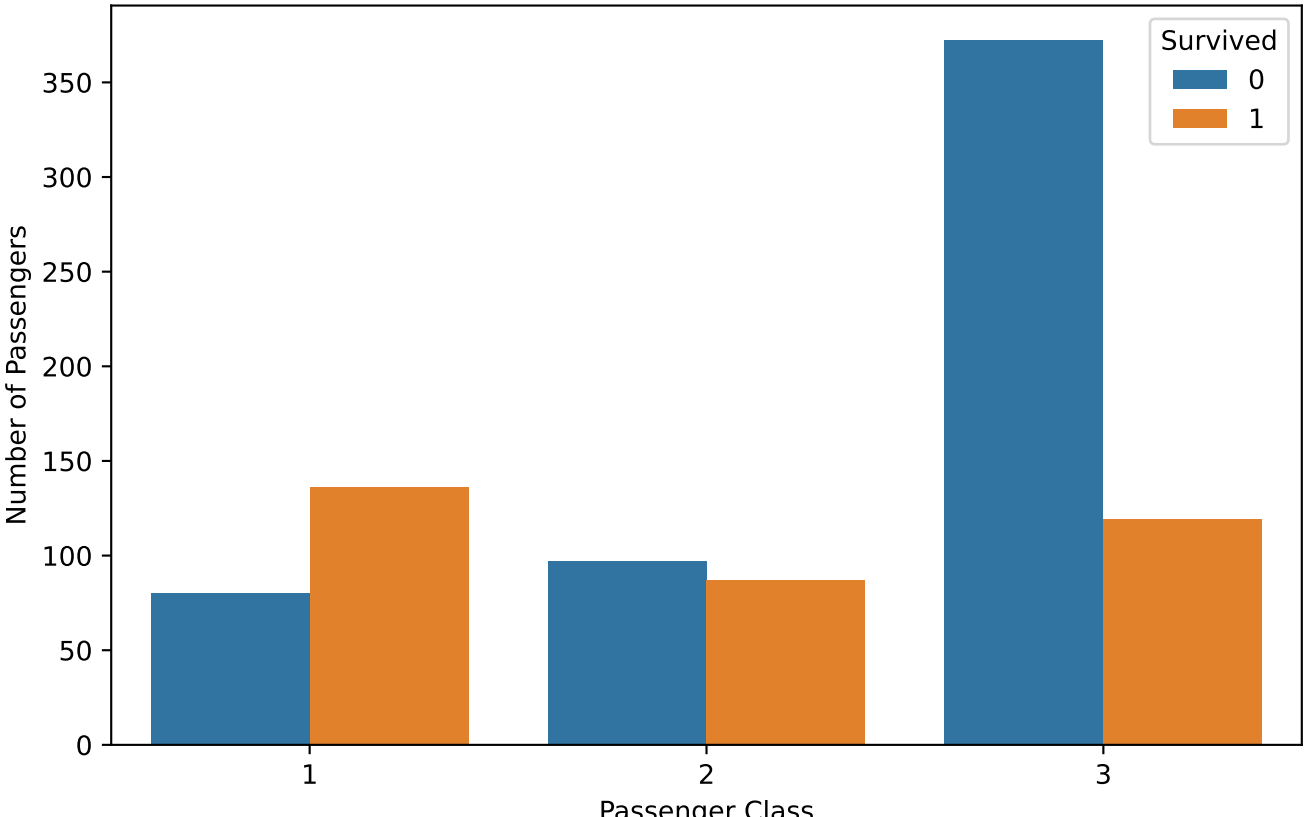
Key findings
- More passengers did not survive than survived.
- Age values vary across passengers.
- Survival outcomes differ by passenger class.
- Survival outcomes also differ by gender.
- The charts help explore numerical relationships and identify potential outliers.

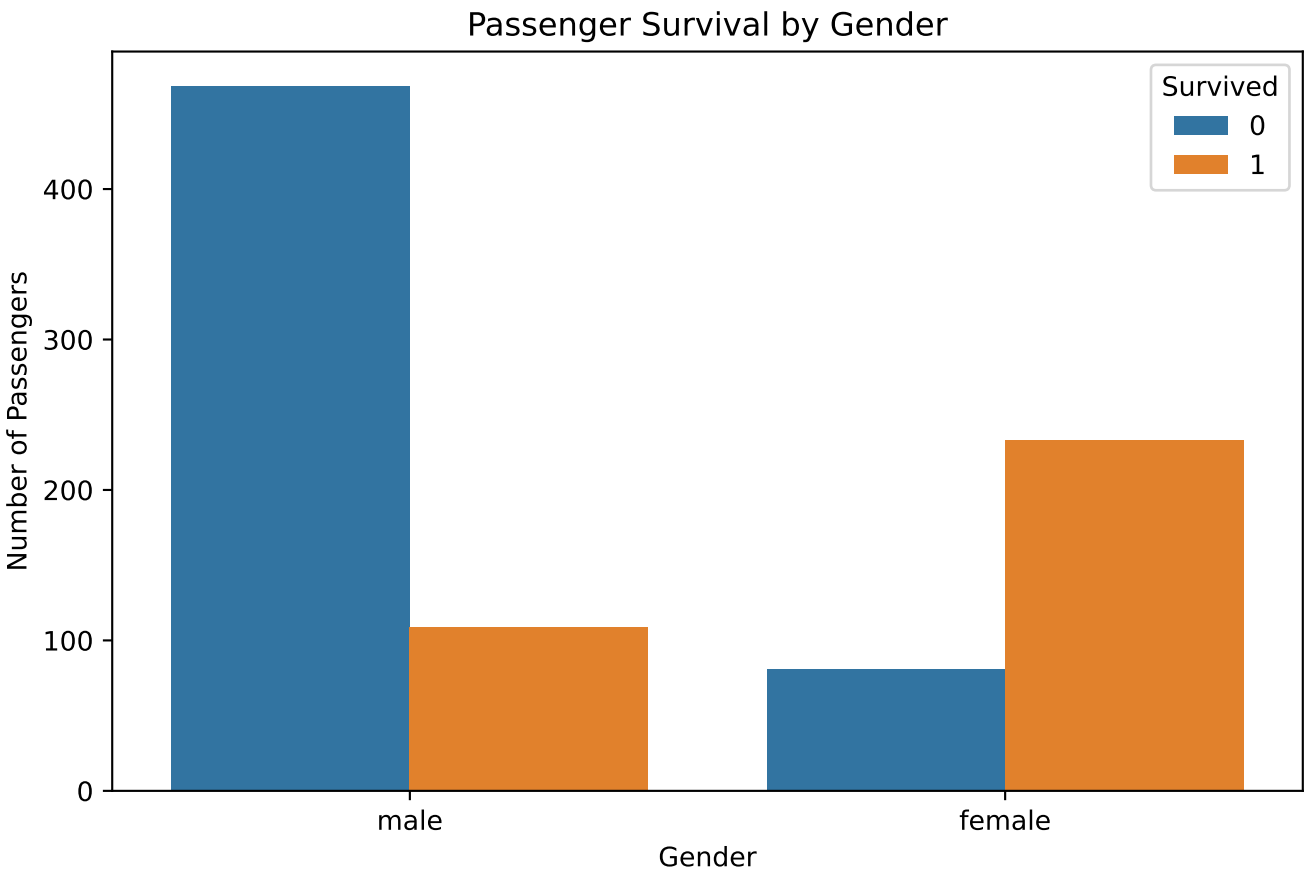
Conclusion
Pandas was used for statistical exploration, while Matplotlib and Seaborn were used to create visualizations. The analysis highlights data quality, distributions, and patterns in Titanic passenger survival.

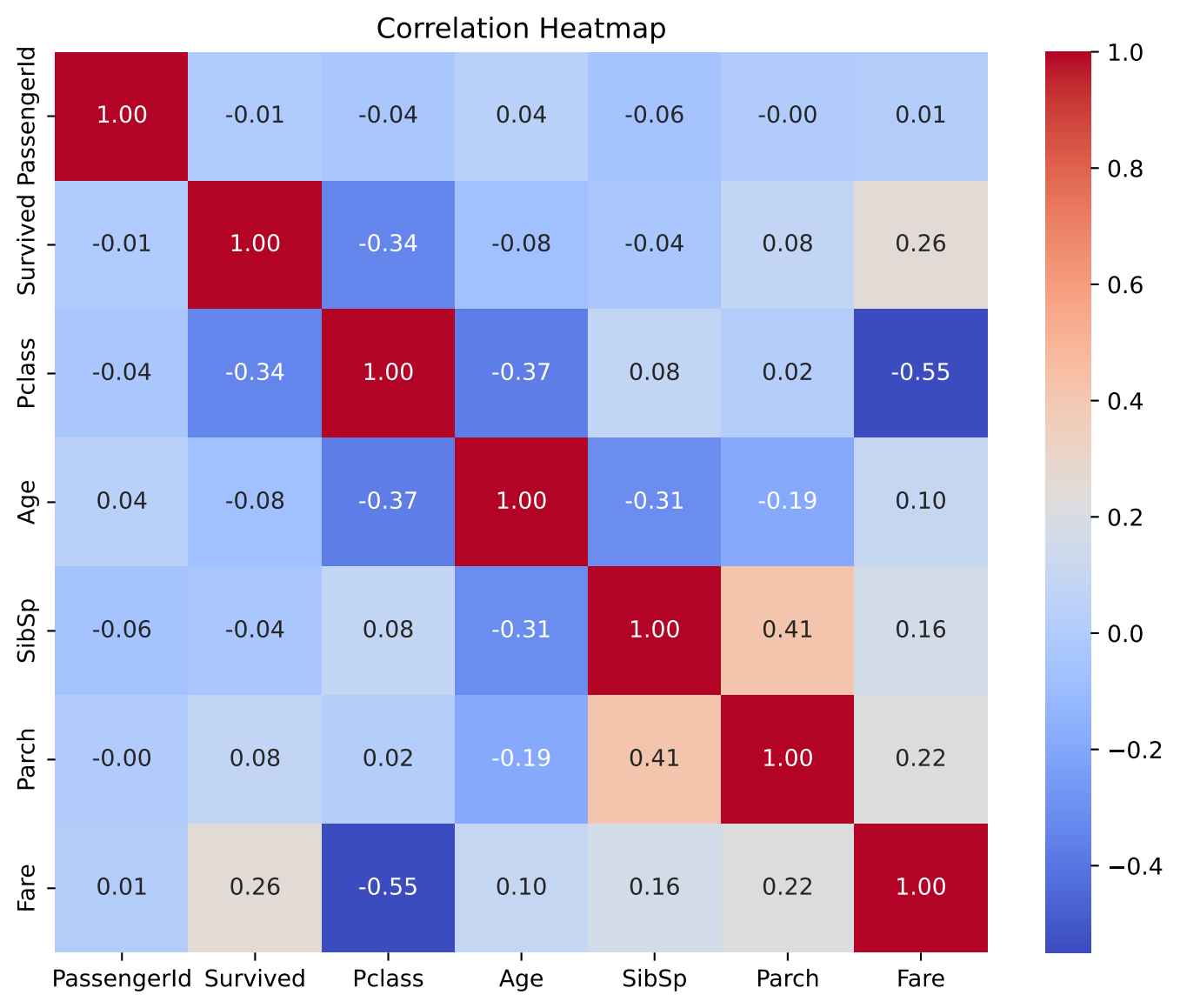
Note: Findings describe associations in this dataset and do not establish cause and effect.


Passenger Survival Count
Number of Passengers
0
100
200
300
400
500
0
1
Survived (0 = No, 1 = Yes)


Distribution of Passenger Ages
Number of Passengers
Age
0
20
40
60
80
100
0
10
20
30
40
50
60
70
80


Passenger Survival by Class
Survived
0
1
Number of Passengers
0
50
100
150
200
250
300
350
1
2
3
Passenger Class


Passenger Survival by Gender
Survived
0
1
Number of Passengers
0
100
200
300
400
male
female
Gender


Correlation Heatmap
PassengerId
Survived
Pclass
Age
SibSp
Parch
Fare
1.00 -0.01 -0.04 0.04 -0.06 -0.00 0.01
-0.01 1.00 -0.34 -0.08 -0.04 0.08 0.26
-0.04 -0.34 1.00 -0.37 0.08 0.02 -0.55
0.04 -0.08 -0.37 1.00 -0.31 -0.19 0.10
-0.06 -0.04 0.08 -0.31 1.00 0.41 0.16
-0.00 0.08 0.02 -0.19 0.41 1.00 0.22
0.01 0.26 -0.55 0.10 0.16 0.22 1.00
PassengerId
Survived
Pclass
Age
SibSp
Parch
Fare
1.0
0.8
0.6
0.4
0.2
0.0
−0.2
−0.4